

Shared Autonomy of a Robotic Manipulator for Grasping Under Human Intent Uncertainty Using POMDPs

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1 Introduction

- Shared autonomy (SA) typically relies on **passive observation** of user joystick inputs to predict intent.
- However, cluttered environments and multiple valid grasps in real-world scenarios make this challenging.
- To provide effective assistance, SA systems should be able to **actively gather information from the human when uncertainty is high** and move towards the goal when uncertainty is low.

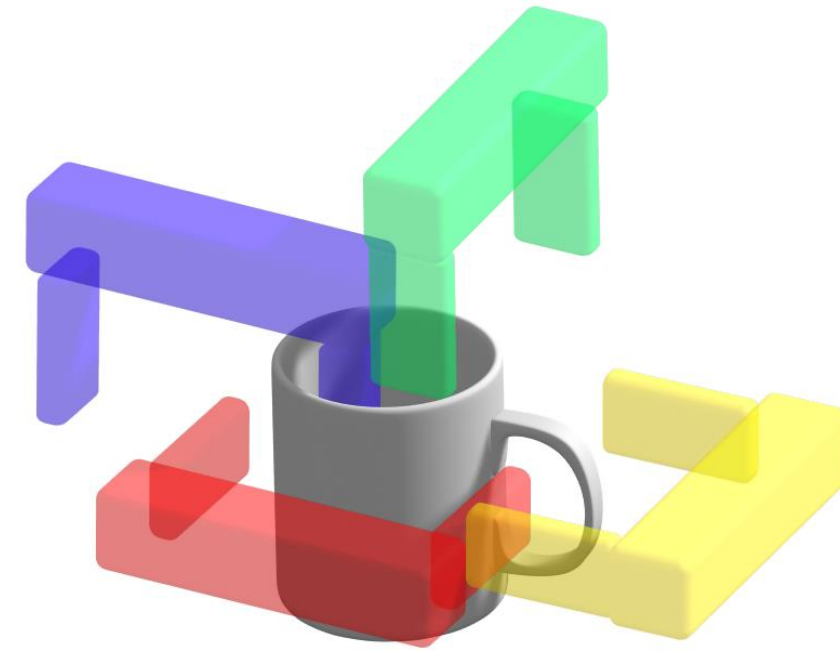


Figure 1: Multiple potential grasping strategies for the same object, each represented by a different color indicating a distinct grasp orientation.

Contributions:

- Demonstrating the need for active information-gathering in SA.
- Modeling SA as a discrete-action POMDP by defining relevant high-level actions.
- User studies comparing our approach against hindsight optimization^[1].

2 Approach

We formulate shared autonomy (SA) as a discrete action Partially Observable Markov Decision Process (POMDP), which offers a principled framework to balance exploration and exploitation.

- On top of *goal-oriented* and *question-asking* actions, we include a *no assist* acting to allow arbitration of control and a *hindsight optimization* action to assist towards a distribution of goals.

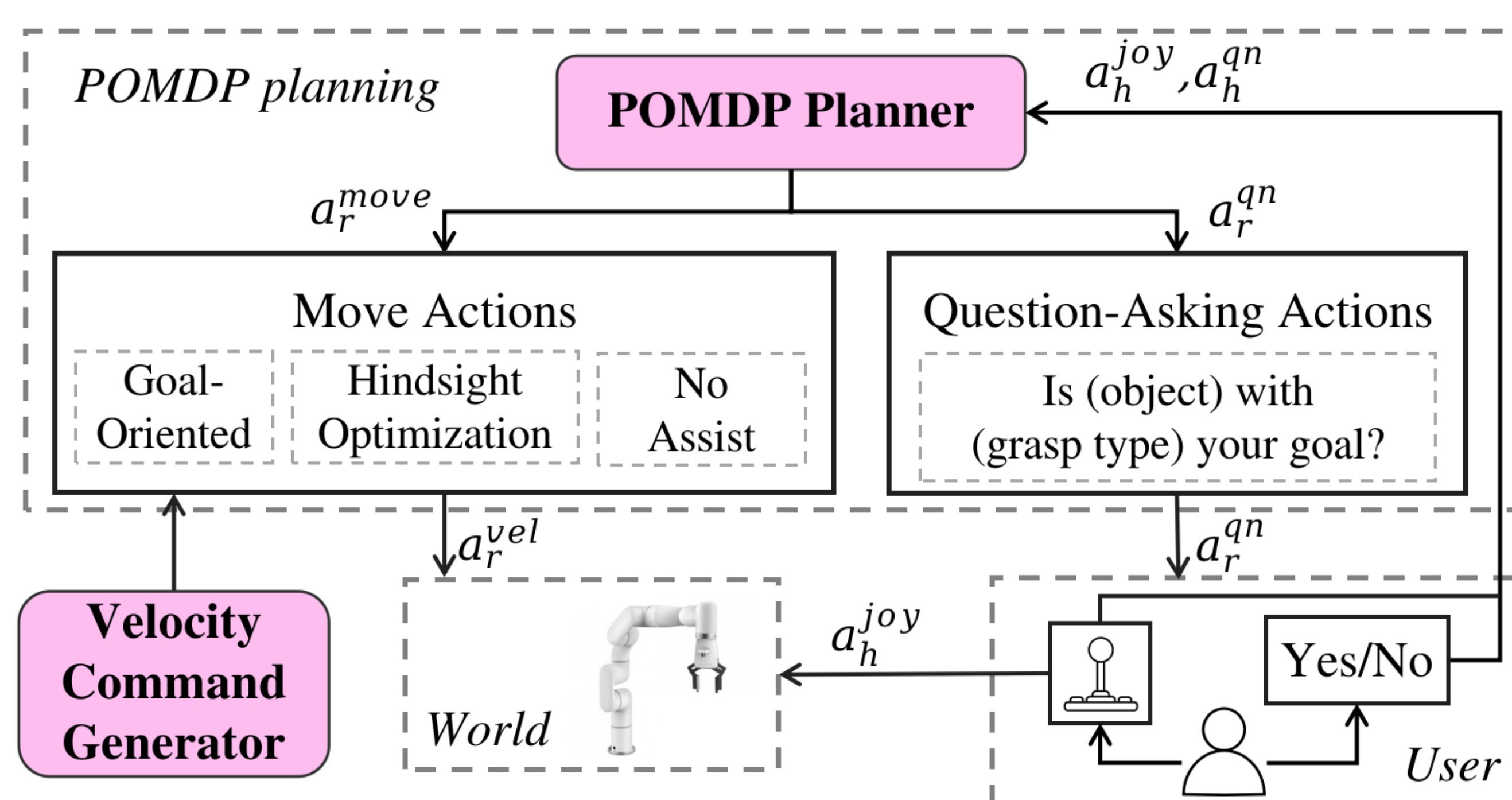


Figure 2: An active information-gathering SA framework using POMDPs. Instead of planning over low-level actions such as moving left or right, we plan over high-level actions, either moving towards a goal or asking questions about a goal.

3 Experiments

- Two user studies were conducted: (1) Simple scene with 18 participants (15 healthy, 2 stroke, and 1 Duchenne muscular dystrophy) (2) Complex scene with 15 healthy participants.
- Participants used a 3-DoF joystick to control the robotic arm and grasp objects with a specific grasp type.
- Baselines: hindsight optimization (HO)^[1], manual teleoperation.



Figure 3: Experimental setups. (Left) Simple scene with four objects. (Right) Complex scene with eleven objects.

4 Results

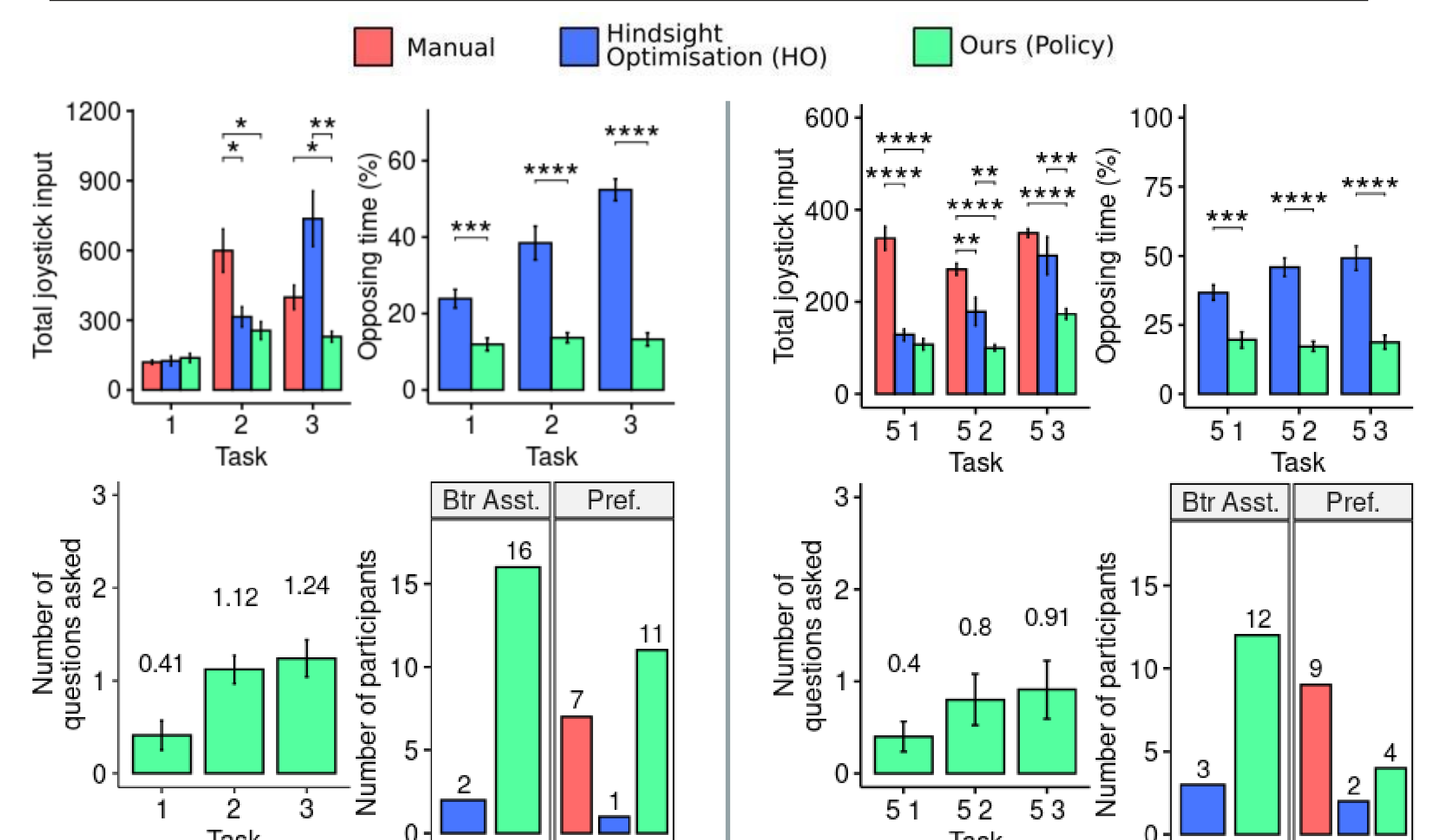


Figure 4: Experimental results. (Left) Simple scene. (Right) Complex scene.

- Our approach outperforms baseline methods, reduces opposing actions and joystick inputs, especially for more difficult tasks involving gripper rotation.
- Our approach effectively reasons when to ask questions, averaging < 2 questions, even in challenging scenarios.
- Our approach was preferred over HO, though some healthy participants still favored manual control for greater autonomy.

5 Conclusion & Future Work

- Our SA framework enhances assistance through active information gathering and by withholding assistance when uncertainty is high, reducing user effort and minimizing conflicts.
- While our user study showed improved performance, the “Yes” and “No” answers remain limited. Future work can consider richer user responses for more intuitive interaction.
- Interestingly, even though the questions asked are meant for information gathering, many participants liked the questions asked as they provided some insights into the robot’s behaviour. Future work can explore additional feedback methods to enhance transparency and user understanding.

For more details, please refer to our paper:



Reference:

[1] Javdani, S., Admoni, H., Pellegrinelli, S., Srinivasa, S. S., & Bagnell, J. A. (2018). Shared autonomy via hindsight optimization for teleoperation and teaming. The International Journal of Robotics Research, 37(7), 717-742.